Simon Zeng

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Education

PhD Jun 2021-Now

University of Bristol, Bristol, UK

I am supervised by Dr Alberto Gambaruto and Dr Tilo Burghardt. We are involved in a NIHR-funded project on Swallowing Rehabilitation. My part of work focuses on Medical Image Processing using Deep Learning and Computational Fluids Dynamics using Meshless method.

Mechanical Engineering Sep 2015-Jun 2020

University of Bristol, Bristol, UK

- **Courses included**: Previous courses - Computer Modelling, Engineering mathematics, Thermofluids, Materials; Ongoing courses – Entrepreneurship, Engineering for International Development, Engineering Design, New power generation.
- **Dissertations**: Wind assessment and Computational Fluids Dynamics simulation using porous media approach;
  Solving Partial Differential Equations using Radial Basis Functions: Image processing using Canny Edge Detection and Surface Reconstruction utilised Poisson Reconstruction, RBF and HRBF

A levels Sep 2013-Jun 2015

John Leggott College, Lincoln, UK

- Math(A*); Advanced math(A); Physics(B); Electronics(B)

Work Experience

Machine Learning Engineering Intern Jul 2020 – Sep 2020

Yaohua Medical Instrument Co.,Ltd, Shandong, China

- **Data Pre-processing:** Normalise and de-noise medical data, remove redundants and duplicates, reduce high dimensionality using PCA
- **Data interpolation:** Utilised different math techniques such as gaussian approximation to interpolate data
- **Data Prediction:** Investigate regression models and normalising methods such as Ridge and Lasso regression to predict data

Co-Founder; Head of Business Development. Jun 2019 – Jan 2020

Erdemy Education Ltd, Bristol, UK

- **Strategic managements:** Make key development decisions with CEO and external consultant employing the knowledge of the current market and competitors.
- **Marketing:** Use efficient strategies to approach Chinese-students market in the UK, Reduce negative net income by 70% in the third month.
- **Relationships:** Help establishing successful collaborations with other companies and organising events

CFD Analysis Intern Jun 2018 – Aug 2018

K8T CFD Analysis Ltd, Bristol, UK

- **CFD analysis:** Designed a variables-controlled case to compared 2D and 3D porous media in ANSYS, compiled a valuable report based on both tests and researches.

- **Post-processing Automation:** Developed a report-generator based on Python, which standardises post-processing report efficiently.

Campus Ambassador Dec 2017 – May 2018
Hino Travel. Ltd, London, UK

- **Campus Advertising:** Promoted products and bundles to clients using both online networking strategy.
- **Customer Services:** Communicated with clients, provided information and took orders

Assistant Researcher Aug 2017 – Sep 2017
Rainbowfish Ocean Technology Co. Ltd, Shanghai, China

- **Sediments Analysis:** Assisted researcher analysing sediments from Mariana Trench, which is useful to investigate the climate over prehistorical period.

Industrial Intern Jul 2015 – Aug 2015
Kingtype High Technology Group, Chengdu, China

- **Assistant:** Assisted at the microchip manufacturing department.

Other Experiences

Founder & Executive President May 2021 – Now
Bristol and West England Chinese Doctoral Association, Bristol, UK
We create this platform with the functionality of Research, Entrepreneurship and Enterprise. Within 3 months, we gathered over 300 PhD students, Postdocs and lectures to join.

President May 2018 – May 2019
Chinese Students' and Scholars' Association, Bristol, UK

- **Administration:** Managed 4 departments and over 40 people to work together efficiently.
- **Events organisation:** Organised over 10 events and activities for more than 2000 students. Increased social media followers by 204%.
- **Relations:** Invited Lord Mayor of Bristol and Sectary from the Embassy of China to the events. Successfully attracted over £8000 sponsorships.

Mealtime volunteer Feb 2017 – Jun 2018
Bristol Royal Infirmary, Bristol, UK

Dog walking volunteer Mar 2017 – Sep 2017
Holly Hedge Animal Sanctuary, Bristol, UK

Certifications and Awards

Bristol Plus Award May 2020
Issued by University of Bristol

Google Analytics Sep 2019
Issued by Google

Award of Excellence for Society President Mar 2019
Awarded by Embassy of China, London

Team of the year 2017 Dec 2017
Awarded by Bristol Royal Infirmary

First Aid Certificate Jan 2017
Issued by St John Ambulance

Math Olympics Bronze Award May 2014
Awarded by John Leggott College

Skills

Technical

CFD: Proficient with STAR CCM+; Ansys
CAD: Proficient with Inventor; AutoCad, Rhinos, Simens NX11
FEA: Proficient with Abaqus
Adobe: Premiere pro cc; After effect; Photoshop

Language

English-Fluent; Chinese-Native
Turkish-Basic

Computational language

Python: basic data analysis, image processing and GUI design
Matlab
Latex
HTML